

Incorporating wellbeing into general factor models: a more complete mental state?

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1 Background

General factor modelling of mental health has gained traction in the last two decades, perhaps best exemplified in research on a general factor of psychopathology or the ‘p factor’^{1,2}. Such general factor models have commonly included a general factor dimension, an internalizing dimension and an externalizing dimension³⁻⁴.

We propose adding wellbeing⁵ as another dimension to such general factor models.

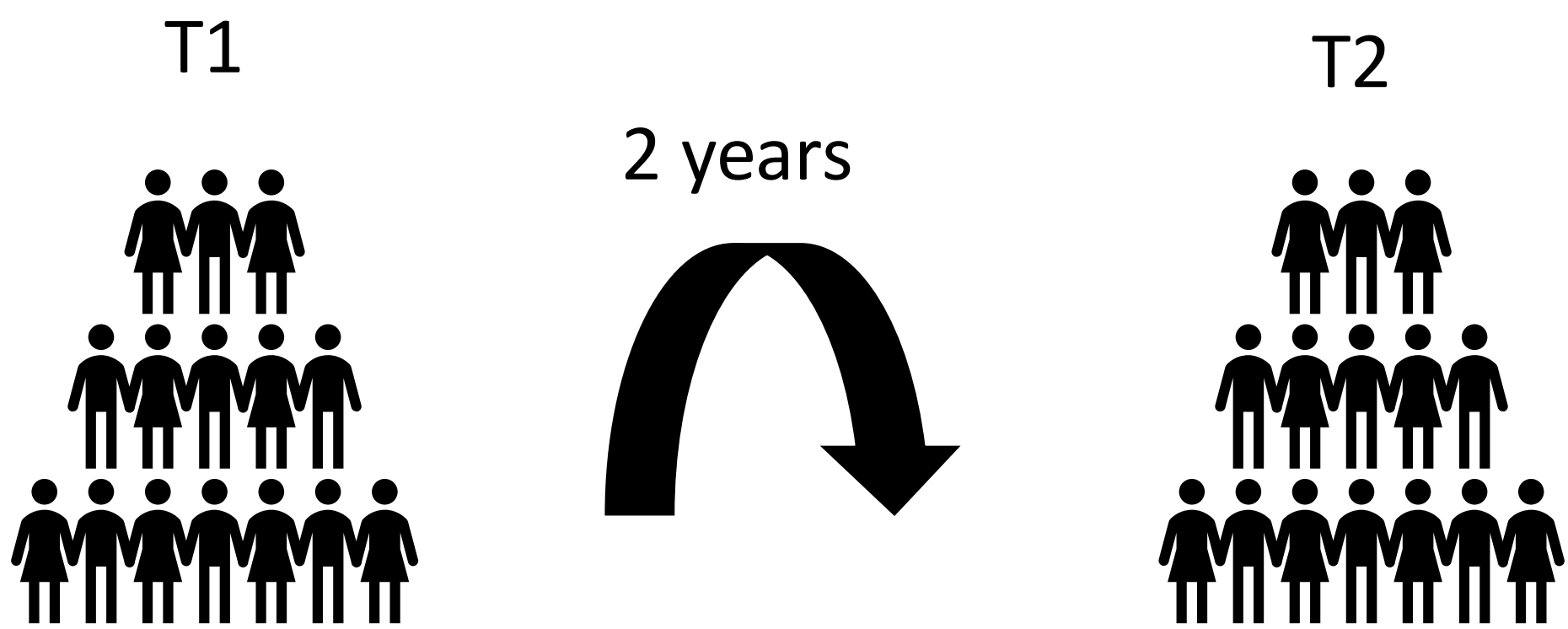
Including wellbeing in general factor models along with psychopathology could have the advantages of:

- (1) representing a more nuanced mental state that indexes shared aspects of both psychopathology and wellbeing
- (2) better accounting for individuals who may cycle between high psychopathology and low wellbeing states and
- (3) being useful in predicting future functional impairment.

We call such a general factor the complete mental state factor (or c factor) to conceptually distinguish it from the p factor.

2 Methods

15258 adolescents enrolled in the HeadStart programme⁶ in UK



Strengths and Difficulties Questionnaire (Psychopathology)
Short Warwick Edinburgh Mental Wellbeing Scale (Wellbeing)
SDQ Impact Supplement (Impairment)

We imputed missing data using multiple imputation by chained equations^{7,8} and used Structural Equation Modelling in R⁹ to estimate measurement and structural models (Figure 1).

3 Results

Measurement Models

CFA Model	Chi-squared	CFI	TLI	RMSEA	SRMR
Bifactor c	p < 0.001	0.948	0.931	0.056	0.049
Correlated factors	p < 0.001	0.922	0.908	0.065	0.064
Bifactor c with method	p < 0.001	0.959	0.945	0.050	0.045
Bifactor p	p < 0.001	0.971	0.949	0.054	0.042
Unidimensional with method	p < 0.001	0.927	0.903	0.067	0.059

Predefined thresholds: CFI & TLI: Acceptable >0.90, Excellent > 0.95;
RMSEA: Acceptable < 0.08, Excellent < 0.06; SRMR < 0.08

Structural Models

Mediation Model	Fit statistics	Path (see Figure 1f)	Pooled std estimate	Pooled std SE
Correlated factors	CFI: 0.913	fut_imp ~ INT	0.14*	0.013
	TLI: 0.900	con_imp ~ INT	0.36*	0.010
	RMSEA: 0.048	fut_imp ~ EXT	0.07*	0.013
	SRMR: 0.056	con_imp ~ EXT	0.21*	0.011
		fut_imp ~ WB	-0.08*	0.012
Bifactor c with method	CFI: 0.939	con_imp ~ WB	-0.14*	0.010
	TLI: 0.925	fut_imp ~ con_imp	0.21*	0.011
	RMSEA: 0.042	fut_imp ~ c	0.31*	0.019
	SRMR: 0.039	con_imp ~ c	0.59*	0.010
		fut_imp ~ con_imp	0.12*	0.017
Bifactor p	CFI: 0.915	fut_imp ~ p	0.23*	0.014
	TLI: 0.884	con_imp ~ p	0.53*	0.009
	RMSEA: 0.050	fut_imp ~ con_imp	0.22*	0.011
	SRMR: 0.036			

fut_imp: future impairment; INT: internalizing factor; EXT: externalizing factor; WB: Wellbeing factor; c: complete mental state factor; con_imp: concurrent impairment; p: p factor

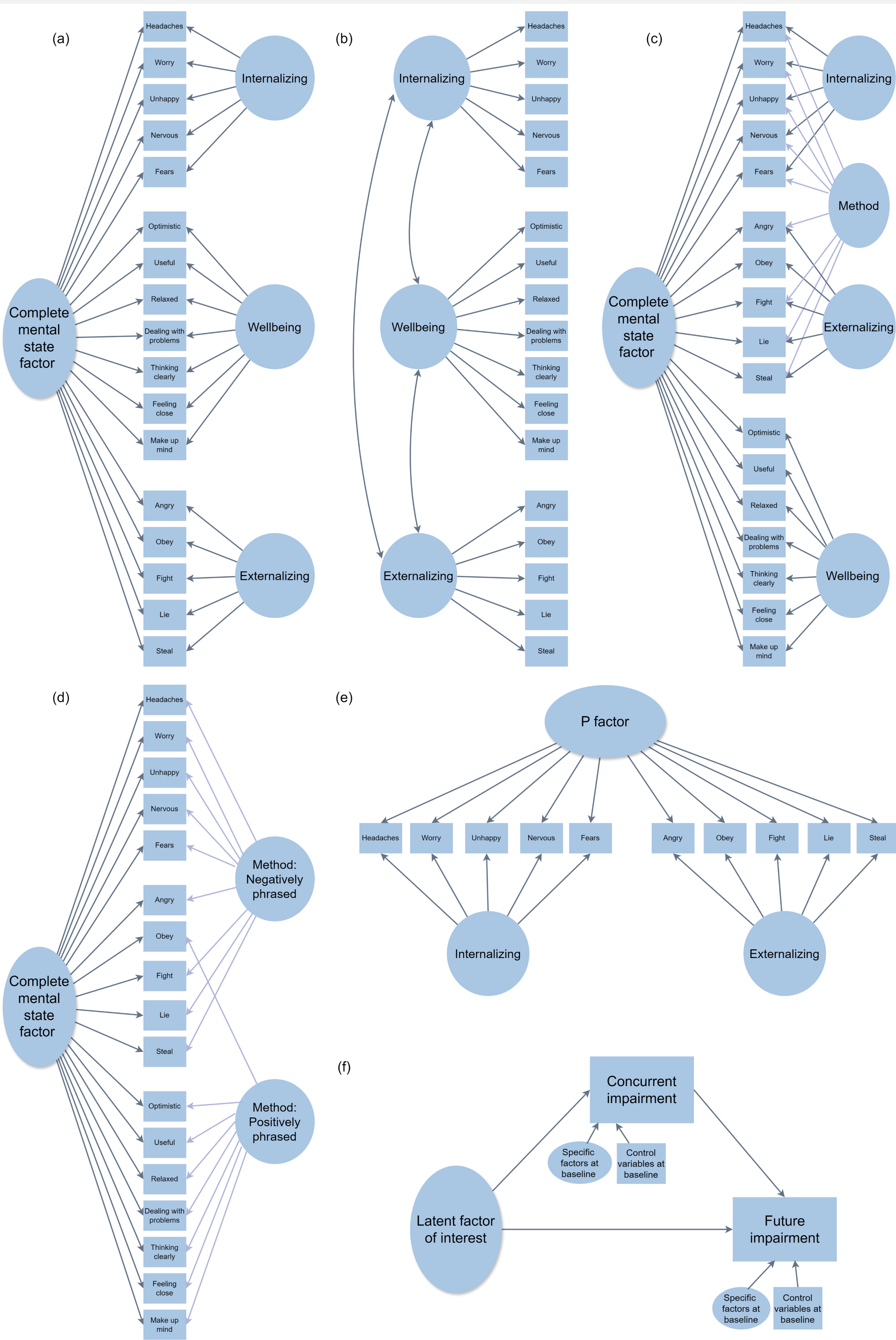


Figure 1. Measurement models. (a) Bifactor c model (b) Correlated factors model (c) Bifactor c with method model (d) Unidimensional model (e) P factor model Structural models. (f) Partial mediation model. The exposure variable is the latent factor of interest (p factor or c factor or specific factors).

4 Discussion & Conclusion

Although both the bifactor p and bifactor c measurement models fit the data, only the bifactor c structural model fits the data. The correlated factors model fits the data but has lower fit indices. Hence, the incorporation of wellbeing into general factor models may be useful for predicting future impairment.

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References
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